ISO 26262 Part 3 - Item Definition

System: Wiper System

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1. Introduction

Clause: 5.1, 5.2

This document's purpose is to define the 'Wiper System' as the 'Item' according to ISO 26262-3:2018. It will specify the Wiper System's role within the vehicle system architecture and detail its relevance to functional safety, focusing on potential hazards and safety goals related to its operation.

2. Requirements of the Item

Clause: 5.4.1

2.1 Legal Requirements, National and International Standards

Clause: 5.4.1 a)

Here's a list of legal, national, and international standards directly applicable to the wiper system, with their relevance to functional safety:

***UNECE Regulation No. 43 (Uniform provisions concerning the approval of safety glazing materials and their installation on vehicles):*

- **Relevance:** While not directly about the wiper system's function, it mandates the use of safety glazing materials (windshields) that the wiper system interacts with. This indirectly impacts safety by ensuring the windshield can withstand wiper operation without cracking or failing, which could impair visibility.

***UNECE Regulation No. 79 (Uniform provisions concerning the approval of vehicles with regard to steering equipment):*

- **Relevance:** Indirectly relevant. While primarily focused on steering, it addresses visibility requirements. If the wiper system fails to clear the windshield adequately, it could impair visibility and indirectly affect steering safety.

***ISO 26262 (Road vehicles — Functional safety):*

- **Relevance:** This is the primary standard for functional safety in automotive systems. It applies to the wiper system if it is electronically controlled or interacts with other safety-critical systems (e.g., automatic emergency braking, rain sensors). ISO 26262 mandates:
- Hazard analysis and risk assessment of the wiper system.
- Safety requirements specification (SRS) for the wiper system.
- Design, implementation, and verification of the wiper system to mitigate identified hazards.
- Management of safety-related development activities.
- Assessment of the wiper system's safety lifecycle.

***SAE J903 (Windshield Wiper Systems):*

- **Relevance:** This standard specifies performance requirements for windshield wiper systems, including wiping area, wiping speed, and blade pressure. While not a functional safety standard in itself, it defines the performance characteristics that are essential for maintaining visibility and thus indirectly contributes to safety. Failure to meet these performance requirements could impair visibility and increase the risk of accidents.

***SAE J1979 (E/E Diagnostic Test Modes):*

- **Relevance:** This standard defines the diagnostic test modes for the wiper system. It is indirectly relevant to functional safety because it ensures that the wiper system can be tested and monitored for proper operation, which is essential for detecting and correcting potential failures that could affect safety.

***IEC 60812 (Analysis techniques for system reliability — Procedure for failure mode and effects analysis (FMEA)):*

- **Relevance:** This standard provides guidance on performing FMEA, which is a key technique used in ISO 26262 to identify potential failure modes of the wiper system and their effects on safety.

***ISO 16750 (Road vehicles — Environmental conditions and testing for electrical and electronic equipment):*

- **Relevance:** This standard specifies environmental conditions and testing for electrical and electronic equipment in road vehicles. It is indirectly relevant to functional safety because it ensures that the wiper system is designed to withstand the environmental conditions it will be exposed to, such as temperature, humidity, and vibration, which can affect its reliability and safety.

Important Considerations:

- **Regional Variations:** While UNECE regulations are widely adopted, specific national implementations may exist.
- **System Complexity:** The extent to which these standards apply depends on the wiper system's complexity (e.g., manual vs. electronically controlled, integration with other systems).
- **OEM Responsibility:** The vehicle manufacturer (OEM) is ultimately responsible for ensuring the wiper system meets all applicable safety requirements.
- **Component Suppliers:** Suppliers of wiper system components (e.g., motors, controllers) must also comply with relevant standards, particularly ISO 26262 if their components are safety-related.

2.2 Functional Behaviour at Vehicle Level

Clause: 5.4.1 b)

Here's a breakdown of the Wiper System's behavior in various operational modes, using precise technical language:

1. Standby Mode

***(1) Triggering Conditions:*

- System power-up initialization sequence completion.
- Absence of any wiper activation request (e.g., from the driver, rain sensor, or automated system).
- No detected fault conditions.

***(2) Primary Functions Performed:*

- Continuous monitoring of input signals (e.g., ignition status, wiper switch position, rain sensor output, system voltage).
- Self-diagnostic routines to verify the integrity of critical components (e.g., motor, park switch, relay).
- Power management to minimize current draw.

***(3) Outputs Produced:*

- None (except for internal diagnostic data).
- Low-power state for electronic control unit (ECU).

***(4) Transition Conditions to Other Modes:*

- **Active Mode:** Wiper switch activation (low, high, intermittent), rain sensor signal exceeding a pre-defined threshold, or automated system request.
- **Fault Mode:** Detection of a critical system malfunction (e.g., motor stall, open circuit, short circuit).
- **Maintenance Mode:** Activation of the maintenance mode switch (if equipped).

2. Active Mode

***(1) Triggering Conditions:*

- Wiper switch position change (low, high, intermittent).
- Rain sensor signal exceeding a pre-defined threshold (for automatic operation).
- Automated system request (e.g., vehicle speed-dependent wiping).

***(2) Primary Functions Performed:*

- Energizing the wiper motor relay based on the selected wiper speed (low, high, intermittent).
- Controlling the wiper motor speed and direction via pulse-width modulation (PWM) or direct voltage application.
- Monitoring the wiper arm position via the park switch.
- Implementing intermittent wipe cycles based on the selected delay time.
- Rain sensor signal processing and interpretation (if equipped).

***(3) Outputs Produced:*

- Energized wiper motor relay.
- PWM signal or voltage applied to the wiper motor.
- Wiper arm movement across the windshield.
- Status signals (e.g., wiper position, motor current).

***(4) Transition Conditions to Other Modes:*

- **Standby Mode:** Wiper switch deactivated, rain sensor signal below the threshold, or automated system request cessation.
- **Fault Mode:** Detection of a critical system malfunction during operation (e.g., motor stall, excessive current draw).
- **Maintenance Mode:** Activation of the maintenance mode switch (if equipped).

3. Fault Mode

***(1) Triggering Conditions:*

- Detection of a critical system malfunction during any operational mode. Examples include:
- Motor stall (excessive current draw).
- Open circuit in the motor or wiring.
- Short circuit in the motor or wiring.
- Park switch failure (failure to signal park position).
- Communication errors with the rain sensor (if equipped).

***(2) Primary Functions Performed:*

- De-energizing the wiper motor relay to prevent further damage.
- Storing the fault code in non-volatile memory.
- Illuminating a warning indicator on the instrument panel (e.g., a wiper symbol).
- Disabling wiper operation.
- Initiating a diagnostic routine to identify the specific fault.

***(3) Outputs Produced:*

- De-energized wiper motor relay.
- Warning indicator activation.
- Fault code stored in memory.
- Diagnostic data (e.g., fault type, timestamp).

***(4) Transition Conditions to Other Modes:*

- **Standby Mode:** After the fault is cleared (e.g., by a repair or reset procedure) and the system is re-initialized.
- **Maintenance Mode:** Not applicable; the system remains in Fault Mode until the fault is resolved.

4. Maintenance Mode

***(1) Triggering Conditions:*

- Activation of the maintenance mode switch (if equipped). This may involve a specific sequence of switch operations or a dedicated button.

***(2) Primary Functions Performed:*

- Positioning the wiper arms in a specific service position (e.g., vertical, away from the windshield).
- Disabling the automatic wiper function (if equipped).
- Allowing manual movement of the wiper arms for cleaning or blade replacement.

***(3) Outputs Produced:*

- Wiper motor energized to move the arms to the service position.
- Disabled automatic wiper functionality.
- Status signal indicating maintenance mode active.

***(4) Transition Conditions to Other Modes:*

- **Standby Mode:** Deactivation of the maintenance mode switch. The system returns to Standby Mode and awaits further activation requests.
- **Fault Mode:** Not applicable; the system will not transition to Fault Mode unless a fault is detected during the maintenance mode operation.

2.3 Required Quality, Performance, and Availability

Clause: 5.4.1 c)

Okay, let's define some quantitative performance requirements relevant to safety, along with their safety rationale. I'll provide examples, and you can adapt them to your specific system or application. Remember that these are illustrative; the actual values will depend on the specific hazards and risk assessment of your system.

Example Scenario: Automated Emergency Shutdown System for a Chemical Reactor

1. Accuracy:

- **Requirement:** Temperature Sensor Accuracy: ±0.5°C
- **Safety Rationale:** Accurate temperature readings are crucial for detecting overheating conditions. This accuracy ensures that the system can reliably identify a temperature excursion (deviation from normal operating temperature) before it reaches a critical threshold that could lead to a runaway reaction or explosion. The ±0.5°C tolerance allows for some sensor drift and measurement error while still providing sufficient margin for safe operation.

2. Response Time:

- **Requirement:** Emergency Shutdown Actuation Time (from detection of a fault to complete shutdown): ≤ 500 ms
- **Safety Rationale:** A rapid response time is essential to mitigate the consequences of a hazardous event. In this scenario, a fast shutdown prevents the reactor from exceeding critical pressure or temperature limits. The 500 ms target allows for:
- Sensor signal processing and fault detection.
- Control system decision-making.
- Actuation of valves to stop reactant flow and initiate cooling.
- This rapid response minimizes the potential for a catastrophic event.

3. Availability:

- **Requirement:** Emergency Shutdown System Availability: ≥ 99.99%
- **Safety Rationale:** High availability ensures that the safety system is functional and ready to respond to a hazardous event when needed. A 99.99% availability translates to a very small amount of downtime per year (approximately 5 minutes). This high availability is achieved through:
- Redundancy (e.g., multiple sensors, controllers, and actuators).
- Regular testing and maintenance.
- Fail-safe design (e.g., valves that automatically close upon loss of power).
- This minimizes the probability that the safety system will be unavailable when a dangerous situation arises.

4. Reliability:

- **Requirement:** Emergency Shutdown System Mean Time Between Failures (MTBF): > 50,000 hours
- **Safety Rationale:** High reliability means the system is less likely to fail during its operational lifetime. A high MTBF indicates that the system is robust and can operate for a long time without requiring repair or replacement. A MTBF of > 50,000 hours (approximately 5.7 years) means that, on average, the system is expected to operate for more than 5 years before a failure occurs. This is achieved through:
- Component selection (using high-quality, reliable components).
- Derating of components (operating components below their maximum ratings).
- Regular preventative maintenance.
- This reduces the risk of the safety system failing to function when it is needed.

5. Other Considerations (Examples):

- **Requirement:** Pressure Relief Valve Flow Capacity: Sufficient to handle the maximum credible overpressure scenario.
- **Safety Rationale:** The valve must be able to safely vent excess pressure to prevent a vessel rupture.
- **Requirement:** Sensor Drift Rate: < 0.1°C per month.
- **Safety Rationale:** Limits the amount of error that can accumulate over time, ensuring that the system remains accurate and reliable.
- **Requirement:** Power Supply Backup Time: ≥ 1 hour.
- **Safety Rationale:** Provides sufficient time for a safe shutdown in the event of a power outage.

Key Considerations for Defining Your Requirements:

- **Hazard Analysis:** Thoroughly analyze the potential hazards associated with your system. Identify the specific failure modes and their consequences.
- **Risk Assessment:** Evaluate the likelihood and severity of each hazard. This will help you prioritize safety requirements.
- **Safety Integrity Level (SIL):** Determine the required SIL for your safety functions. This will influence the stringency of your performance requirements. (SILs are defined by standards like IEC 61508 and IEC 61511).
- **Standards and Regulations:** Comply with all applicable safety standards and regulations.

- **Redundancy and Diversity:** Consider using redundant and diverse safety systems to improve reliability and reduce the risk of common-cause failures.
- **Testing and Validation:** Implement rigorous testing and validation procedures to ensure that your safety system meets its performance requirements.

How to Apply This to Your System:

1. **Identify the Hazards:** What are the potential dangers associated with your system?

2. **Define Safety Functions:** What actions must be taken to mitigate each hazard? (e.g., "Shut down the pump," "Close the valve," "Activate the alarm.")

3. **Determine Critical Parameters:** What parameters need to be monitored to trigger the safety functions? (e.g., temperature, pressure, flow rate, position)

4. **Set Performance Requirements:** For each critical parameter and safety function, define the accuracy, response time, availability, and reliability requirements. Base these on the hazard analysis, risk assessment, and any applicable standards.

5. **Document and Justify:** Clearly document all requirements and provide a safety rationale for each one.

By following these steps, you can develop a robust set of quantitative performance requirements that will help ensure the safety of your system. Remember to consult with safety experts and engineers throughout the process.

2.4 Constraints Regarding the Item

Clause: 5.4.1 d)

Here's a breakdown of constraints affecting a Wiper System, categorized as requested, along with operating ranges and consequences of violation:

1. Environmental Constraints:

Constraint	Operating Range	Consequence of Violation

2.5 Potential Consequences of Behavioural Shortfalls

Clause: 5.4.1 e)

Here's a breakdown of critical wiper system functions, potential hazards, harms, affected parties, and safety goals:

| Critical Function | Failure/Malfunction Scenario | Hazardous Event | Potential Harm | Affected Parties | Potential Safety Goal |

|---|---|---|---|---|---|

| **Wiper Blade Movement (Wiping)** | Blades fail to move, move erratically, or streak/smear water/debris. | Reduced visibility in inclement weather (rain, snow, sleet, fog, spray). | Difficulty seeing the road, other vehicles, pedestrians, obstacles. Increased risk of: - Lane departure - Delayed reaction to hazards - Collision | Driver, Passengers, Pedestrians, Cyclists, Other Motorists | **Maintain Clear Visibility:** Ensure the driver has adequate visibility in all weather conditions to perceive hazards and control the vehicle. |

| **Wiper Speed Control (Variable/Intermittent)** | Wiper speed is too slow for the amount of precipitation, or the intermittent function fails to clear the windshield effectively. | Reduced visibility in inclement weather. | Difficulty seeing the road, other vehicles, pedestrians, obstacles. Increased risk of: - Delayed reaction to hazards - Collision | Driver, Passengers, Pedestrians, Cyclists, Other Motorists | **Provide Adaptive Visibility:** Ensure the wiper system adjusts to the intensity of precipitation to maintain a clear view of the road. |

| **Washer Fluid Delivery** | Washer fluid fails to spray, sprays erratically, or sprays in the wrong direction. | Inability to clear road grime, salt, or other debris from the windshield. | Reduced visibility, especially when combined with glare or low sun angles. Increased risk of: - Delayed reaction to hazards - Collision | Driver, Passengers, Pedestrians, Cyclists, Other Motorists | **Ensure Effective Cleaning:** Provide a reliable means of cleaning the windshield to remove debris and maintain visibility. |

| **Wiper Park Position** | Wipers fail to park correctly (e.g., obstruct driver's view, block sensors). | Obstructed vision, potential damage to wipers or windshield. | Reduced visibility, potential for damage to the wiper system. Increased risk of: - Delayed reaction to hazards - Collision | Driver, Passengers, Pedestrians, Cyclists, Other Motorists | **Ensure Safe Stowage:** Ensure the wipers park in a position that does not obstruct the driver's view or interfere with vehicle operation. |

| **Wiper Motor/Mechanism Failure** | Complete failure of the wiper system (motor seizes, linkage breaks). | Complete loss of visibility in inclement weather. | Inability to see the road, other vehicles, pedestrians, obstacles. Extremely high risk of: - Collision - Loss of control | Driver, Passengers, Pedestrians, Cyclists, Other Motorists | **Provide Redundant Visibility (if possible):** Ensure the driver can maintain some level of visibility in the event of a complete

wiper failure (e.g., through defogging, emergency stopping). **OR Ensure System Reliability:** Design the wiper system to be highly reliable and resistant to failure. |

| **Sensor Integration (Rain Sensors)** | Rain sensor malfunctions, triggering wipers inappropriately (e.g., when it's not raining) or failing to activate them when needed. | Distraction from unnecessary wiper operation, or reduced visibility in rain. | Driver distraction, reduced visibility. Increased risk of: - Driver distraction - Delayed reaction to hazards - Collision | Driver, Passengers, Pedestrians, Cyclists, Other Motorists | **Provide Reliable Automation:** Ensure the rain sensor accurately detects precipitation and activates the wipers appropriately. |

2.6 Capabilities of the Actuators (or Assumed Capabilities)

Clause: 5.4.1 f)

Okay, let's break down the actuation capabilities of a Wiper System, focusing on the aspects relevant to Hazard Analysis and Risk Assessment (HARA) and controllability/severity assessments. We'll quantify these capabilities, keeping in mind the need for safety-critical performance.

1. Maximum Response Time for Safety Actions

This is the most critical aspect for safety. The response time dictates how quickly the system can react to a hazard, such as obscured vision. We need to consider both the *detection* of the hazard (e.g., rain sensor activation) and the *actuation* of the wipers.

- **Scenario:** Sudden heavy rain obscuring vision.
- **Safety Action:** Wiper blades must move to clear the windshield.

***Quantification:*

- **Detection to Actuation Delay (Total Response Time):** This is the sum of several delays:
- **Rain Sensor Detection Time:** This is the time it takes for the rain sensor to detect the rain and signal the wiper control module. This should be *very fast* to minimize the time the driver's vision is impaired. **Target: ≤ 100 ms** (This is a common target, but can be improved with advanced sensors).
- **Control Module Processing Time:** The time for the wiper control module to process the rain sensor signal and determine the appropriate wiper speed. **Target: ≤ 20 ms** (This should be very fast).

- **Motor Start-up Time:** The time it takes for the wiper motor to start moving the blades. This is the most significant delay. **Target: ≤ 200 ms** (This is a critical parameter. Faster is better, but depends on motor design and gear ratio).
- **Blade Travel Time (to clear the critical vision area):** The time for the wiper blades to sweep across the windshield and clear the driver's primary field of view. This depends on the blade length, sweep angle, and motor speed. **Target: ≤ 500 ms** (This is a critical parameter. Faster is better, but depends on blade length, sweep angle, and motor speed).
- **Overall Target (Worst-Case): ≤ 820 ms** (Sum of the above, assuming worst-case conditions). This is a *critical* safety parameter. Any delay here directly impacts the driver's ability to see. This value should be validated through testing and analysis.
- **Important Note:** This is a *system-level* requirement. Each component (sensor, control module, motor) must contribute to meeting this overall target. The system design must allocate time budgets to each component.

2. Force/Torque/Current Limits

These limits are crucial for preventing damage to the wiper system, the windshield, and for ensuring safe operation. They also help in diagnosing faults.

***Force/Torque Limits:*

- **Maximum Blade Force:** The maximum force the wiper blades can exert on the windshield. This is important to prevent damage to the windshield, especially under freezing conditions or when the blades are stuck. **Target: < 50 N** (This is a typical value, but should be determined based on windshield material and wiper blade design). This should be measured at the blade tip.
- **Maximum Motor Torque:** The maximum torque the wiper motor can produce. This is related to the motor's power and gear ratio. **Target: Defined by motor specifications, but should be limited to prevent mechanical damage.** This should be measured at the motor shaft.
- **Stall Torque:** The torque the motor produces when stalled (blades stuck). This is a critical parameter for protection. **Target: Defined by motor specifications, but should be limited to prevent mechanical damage.** This should be measured at the motor shaft.
- **Torque Limiting:** The system should incorporate torque limiting mechanisms (e.g., current limiting in the motor driver) to prevent excessive force.

***Current Limits:*

***Maximum Motor Current (Continuous):** The maximum current the motor can draw continuously without overheating. **Target: Defined by motor specifications.*

***Maximum Motor Current (Peak/Stall):** The maximum current the motor can draw during startup or when stalled. This is typically higher than the continuous current. **Target: Defined by motor specifications, and should be limited by the motor driver.*

***Current Limiting:** The motor driver should have current limiting to protect the motor and wiring. **Target: Set to a value below the motor's maximum current ratings.*

***Quantification and Measurement:*

- These limits should be specified in the system requirements and verified through testing.
- Force/torque can be measured using force gauges or torque sensors.
- Current can be measured using current sensors or by monitoring the voltage drop across a shunt resistor.

3. Diagnostic Coverage of Actuators

Effective diagnostics are essential for detecting faults and ensuring the wiper system functions reliably. This is critical for HARA and safety.

- **Fault Detection Coverage:** The percentage of potential faults that the diagnostic system can detect. This is a key metric for safety.

***Motor Faults:*

- **Open Circuit:** Detection of an open circuit in the motor windings or wiring. **Coverage: Near 100%** (Easy to detect with current monitoring).
- **Short Circuit:** Detection of a short circuit in the motor windings or wiring. **Coverage: Near 100%** (Easy to detect with current monitoring).
- **Stall:** Detection of the motor being stalled (blades stuck). **Coverage: 100%** (Current monitoring and/or position feedback).
- **Overload:** Detection of excessive current draw. **Coverage: 100%** (Current monitoring).

***Motor Speed Failure:** Detection of the motor not reaching the commanded speed. **Coverage: High, using speed feedback (e.g., from a hall effect sensor) or by monitoring the time it takes to complete a sweep.*

***Wiring Faults:*

- **Open Circuit:** Detection of open circuits in the wiring harness. **Coverage: Near 100%** (Continuity checks).
- **Short Circuit:** Detection of short circuits in the wiring harness. **Coverage: Near 100%** (Continuity checks and current monitoring).
- **Ground Fault:** Detection of a short to ground. **Coverage: Near 100%** (Insulation resistance checks).

***Control Module Faults:*

- **Internal Faults:** Detection of internal faults in the control module (e.g., processor errors, memory errors). **Coverage: Varies depending on the module's design, but should be as high as possible.** Use of watchdog timers, memory checks, and other diagnostic features.
- **Communication Errors:** Detection of communication errors between the control module and other ECUs (e.g., rain sensor, body control module). **Coverage: 100%** (Communication protocols with error detection).

***Actuator Position Feedback:*

***Position Feedback:** If the system uses position feedback (e.g., from a hall effect sensor), the diagnostics should verify the accuracy and validity of the position data. **Coverage: High, using plausibility checks and range checks.*

***Diagnostic Implementation:*

- **Continuous Monitoring:** The system should continuously monitor the motor current, voltage, and other relevant parameters.
- **Error Codes:** The system should generate specific error codes to identify the type of fault.
- **Fault Reporting:** The system should report faults to the driver (e.g., through a warning light) and/or to other ECUs.
- **Fail-Safe Mechanisms:** The system should implement fail-safe mechanisms to mitigate the effects of faults (e.g., stopping the wipers, switching to a default speed).

***Diagnostic Coverage Assessment:*

- **Fault Tree Analysis (FTA):** Use FTA to identify potential failure modes and assess the diagnostic coverage for each mode.
- **Failure Mode and Effects Analysis (FMEA):** Use FMEA to analyze the effects of each failure mode and determine the severity and detectability.

- **Testing:** Perform rigorous testing to verify the diagnostic coverage and ensure that the system can detect and respond to faults effectively.

Connecting to HARA and Controllability/Severity:

- **HARA:** The quantified values above directly support the HARA process. They provide the data needed to:
- **Identify Hazards:** e.g., Obstructed vision due to wiper failure.
- **Assess Risks:** By considering the frequency of the hazard, the severity of the potential harm (e.g., accident), and the controllability of the hazard.
- **Determine Safety Requirements:** The response time, force limits, and diagnostic coverage are all critical safety requirements derived from the HARA.
- **Controllability:** The response time and diagnostic coverage directly impact controllability. A fast response time and good diagnostic coverage increase the driver's ability to control the vehicle and avoid an accident.
- **Severity:** The severity of a hazard is directly related to the performance of the wiper system. A slow response time or a failure to clear the windshield can significantly increase the severity of an accident.

In Summary:

The quantification of these actuation capabilities is essential for designing a safe and reliable wiper system. The values provided are examples and should be tailored to the specific application and vehicle requirements. Thorough testing and analysis are crucial to validate these values and ensure that the system meets the required safety standards. Remember to document all values and testing results for traceability and compliance.

3. Boundary of the Item, Interfaces, and Interaction Assumptions

Clause: 5.4.2

3.1 Elements of the Item

Clause: 5.4.2 a)

Here's a breakdown of the hardware and software elements within the scope of a Wiper System, along with explicit exclusions:

Hardware (Within Scope):

- **ECU (Electronic Control Unit):** The central processing unit that controls the wiper system's operation.

***Sensors:*

- **Rain Sensor:** Detects the presence and intensity of rain on the windshield.
- **Park Position Sensor:** Detects the wiper blade's parked position.
- **Vehicle Speed Sensor (VSS):** Used to adjust wiper speed based on vehicle speed (sometimes).
- **Windshield Temperature Sensor (Optional):** Used to adjust wiper speed or heating based on windshield temperature.

***Actuators:*

- **Wiper Motor:** The electric motor that drives the wiper blades.
- **Wiper Linkage:** The mechanical system that translates the motor's rotary motion into the sweeping motion of the wiper blades.
- **Washer Pump:** The electric pump that delivers washer fluid to the windshield.
- **Washer Nozzles:** The nozzles that spray washer fluid onto the windshield.
- **Wiper Blades:** The rubber blades that wipe the windshield.
- **Heated Wiper Blades (Optional):** If equipped, the heating element within the wiper blades.

Software (Within Scope):

***Modules:*

- **Wiper Control Module:** The core logic that governs wiper operation (e.g., speed selection, intermittent mode, rain sensing logic).
- **Rain Sensing Module:** (If applicable) The software that processes data from the rain sensor.
- **Diagnostics Module:** Software for fault detection and reporting.
- **Communication Module:** Software for communication with other ECUs (e.g., Body Control Module).

***Drivers:*

- **Wiper Motor Driver:** Software that controls the wiper motor's speed and direction.
- **Washer Pump Driver:** Software that controls the washer pump.

- **Sensor Drivers:** Software that interfaces with the sensors (e.g., rain sensor driver, park position sensor driver).

***Libraries:*

- **Sensor Abstraction Libraries:** Libraries that provide a standardized interface for accessing sensor data.
- **Motor Control Libraries:** Libraries that provide functions for controlling the wiper motor.
- **Communication Libraries:** Libraries for communication with other ECUs.
- **Diagnostic Libraries:** Libraries for diagnostic functions.

Excluded (Out of Scope):

- **Battery cells:** Power source for the system.
- **Wiring harness and connectors:** The physical connections between components.
- **Windshield glass:** The surface the wipers clean.
- **Vehicle Body:** The structural components of the vehicle.
- **Human-Machine Interface (HMI) elements:** The physical switches and controls used by the driver (e.g., wiper stalk).
- **Power Distribution Module (PDM) or Fuse Box:** The components that distribute power to the wiper system.
- **Manufacturing equipment:** Tools and machinery used to produce the wiper system components.
- **Testing equipment:** Tools and machinery used to test the wiper system components.
- **Software development tools:** Compilers, debuggers, and IDEs used to create the software.
- **Operating System (OS):** The underlying software that runs on the ECU.
- **Communication protocols:** The specific communication protocols used to communicate with other ECUs (e.g., CAN, LIN).
- **Mechanical components of the vehicle that are not directly part of the wiper system:** e.g., the steering column, the dashboard.

3.2 Assumptions Concerning Effects on the Vehicle

Clause: 5.4.2 b)

Here are some verifiable and realistic assumptions about a vehicle's response to the "Wiper System" outputs, categorized for clarity:

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High-Tech Humanity

I. Wiper System Functionality & Hardware:

- **Assumption 1: Wiper Motor Functionality:** The wiper motor will respond to the wiper system's speed commands (e.g., low, high, intermittent) within a reasonable timeframe (e.g., less than 1 second). This can be verified by testing the motor's response time.
- **Assumption 2: Wiper Blade Contact:** The wiper blades will maintain consistent contact with the windshield surface under normal driving conditions (e.g., speeds up to the vehicle's maximum speed, and in light to moderate rain). This can be verified through visual inspection and testing in various weather conditions.
- **Assumption 3: Washer Fluid Availability:** The washer fluid reservoir has sufficient fluid for at least a few cycles of the washer system. This can be verified by checking the fluid level sensor and the reservoir capacity.
- **Assumption 4: Washer Nozzle Functionality:** The washer nozzles will spray fluid onto the windshield in a predictable pattern when activated. This can be verified by observing the spray pattern.
- **Assumption 5: Wiper System Power:** The wiper system is supplied with sufficient power to operate correctly. This can be verified by checking the voltage and current draw of the system.

II. Driver Interaction & Perception:

- **Assumption 6: Driver Perception:** The driver can visually perceive the windshield's clarity and the effectiveness of the wipers. This is a fundamental assumption for the driver to make informed decisions.
- **Assumption 7: Driver Response to Obstructed Vision:** The driver will react to significantly obstructed vision (e.g., due to heavy rain, snow, or debris) by adjusting their speed, increasing following distance, or pulling over to a safe location. This is a general assumption about driver behavior.
- **Assumption 8: Driver Awareness of Wiper Controls:** The driver is familiar with the location and operation of the wiper controls (e.g., stalk, buttons). This is a prerequisite for the driver to use the system effectively.

III. System Interactions & Safety:

- **Assumption 9: Other ECUs Respect Safety States:** Other Electronic Control Units (ECUs), such as the Anti-lock Braking System (ABS) or Electronic Stability Control (ESC), will not interfere with the wiper system's operation in a way that compromises safety.

For example, the ABS will not disable the wipers during braking. This can be verified through system testing and documentation.

- **Assumption 10: Wiper System Failure Indication:** The vehicle will provide a clear indication (e.g., warning light, message on the instrument cluster) if the wiper system malfunctions significantly (e.g., motor failure, blade detachment). This can be verified by simulating failures and observing the system's response.
- **Assumption 11: Limited Impact on Other Systems:** The wiper system's operation will not significantly impact other vehicle systems (e.g., electrical load, engine performance) under normal operating conditions. This can be verified by monitoring system parameters during wiper operation.
- **Assumption 12: No Interference with Autonomous Systems (If Applicable):** If the vehicle has autonomous driving features, the wiper system will not interfere with the operation of these systems (e.g., by obscuring camera vision). This is crucial for the safe operation of autonomous features.

IV. Environmental Considerations:

- **Assumption 13: Temperature Range:** The wiper system is designed to operate within the vehicle's specified temperature range (e.g., from -30°C to 85°C). This can be verified through testing in extreme temperatures.
- **Assumption 14: Wind Resistance:** The wiper blades are designed to withstand wind resistance at the vehicle's maximum speed. This can be verified through wind tunnel testing or simulations.

Verification Methods:

- **Testing:** Physical testing of the vehicle under various conditions (e.g., rain, snow, different speeds).
- **Simulation:** Using software to simulate the wiper system's behavior and interactions with other systems.
- **Inspection:** Visual inspection of the components and their installation.
- **Documentation Review:** Reviewing the vehicle's specifications, schematics, and software documentation.
- **Failure Mode and Effects Analysis (FMEA):** Analyzing potential failure modes and their effects on the system's operation.

These assumptions are crucial for designing, testing, and validating the wiper system's safety and effectiveness. They provide a framework for understanding how the system should behave and how to ensure its reliable operation. Remember to tailor these assumptions to the specific vehicle and its intended use.

3.3 Functionality Required by Other Items

Clause: 5.4.2 c)

Okay, let's break down the wiper system's interactions with other vehicle systems, focusing on the data/functions it provides. This is a general overview, and specific implementations can vary.

Here's a table outlining the information:

| (1) Recipient System | (2) Data/Function Provided | (3) Update Rate | (4) Safety Relevance |

|---|---|---|---|

| **Body Control Module (BCM)** | *Wiper Motor Speed/Position (e.g., Parked, Low, High, Intermittent)* Wiper Status (On/Off, Active) *Washer Fluid Level (if equipped with a sensor)* Washer Fluid Pump Activation | *Continuous (for speed/position)* Event-driven (On/Off, Washer activation) *Periodic (Fluid Level) |* **Indirect:** Wiper operation is critical for visibility, which is directly related to safety. BCM uses this data for overall vehicle control and user interface. *Indirect:** Washer fluid level is important for maintaining visibility. |

| **Instrument Cluster / Driver Information System (DIS)** | *Wiper Status (On/Off, Speed Setting)* Washer Fluid Level (if equipped) *Rain Sensor Status (if equipped) |* Event-driven (On/Off, Speed changes) *Periodic (Fluid Level)* Event-driven (Rain Sensor) | *Direct:** Provides the driver with critical information about wiper operation and potential visibility issues (low fluid). |

| **Automatic Headlight System / Light Control Module (LCM)** | *Wiper Operation (On/Off, Speed)* Rain Sensor Data (if equipped) | *Event-driven (On/Off, Speed changes)* Continuous (Rain Sensor) | *Indirect:** Can be used to automatically activate headlights in conjunction with wiper operation, improving visibility in poor weather. |

| **Engine Control Unit (ECU) / Powertrain Control Module (PCM)** | *Washer Fluid Pump Activation (for some vehicles) |* Event-driven (Washer activation) | *Indirect:** May be used to temporarily disable or reduce engine load during washer operation to prevent stalling or power loss. |

| **Advanced Driver-Assistance Systems (ADAS) / Autonomous Driving Systems** | *Rain Sensor Data (if equipped)* Wiper Status (On/Off, Speed) *Wiper Position (for some systems) |*

Continuous (Rain Sensor) *Event-driven (On/Off, Speed changes)* Continuous (Position) | *Direct:** Rain sensor data is crucial for ADAS features like automatic emergency braking (AEB), lane keeping assist (LKA), and adaptive cruise control (ACC). Wiper status and position are used to understand visibility conditions and adjust system behavior accordingly. |

| **Climate Control System** | *Rain Sensor Data (if equipped) |* Continuous (Rain Sensor) | *Indirect:** Can be used to automatically activate the defroster or demister based on rain sensor data, improving visibility. |

| **Infotainment System** | *Wiper Status (On/Off, Speed)* Washer Fluid Level (if equipped) | *Event-driven (On/Off, Speed changes)* Periodic (Fluid Level) | *Indirect:** Provides the driver with information about wiper operation and potential visibility issues (low fluid). |

| **Electronic Stability Control (ESC) / Anti-lock Braking System (ABS)** | *Rain Sensor Data (if equipped) |* Continuous (Rain Sensor) | *Indirect:** Can be used to adjust braking behavior based on rain sensor data, improving safety in wet conditions. |

Important Considerations:

- **Rain Sensor Integration:** The presence and sophistication of a rain sensor significantly impact the data provided. More advanced systems provide more detailed information.
- **Vehicle Architecture:** The specific communication protocols (CAN bus, LIN bus, etc.) and data formats will vary depending on the vehicle's architecture.
- **Safety Levels:** The safety relevance is often indirect, as the wiper system's primary function is to improve visibility. However, the data it provides is critical for other systems that directly impact safety.
- **Redundancy:** In safety-critical applications (e.g., ADAS), redundancy might be implemented. For example, multiple rain sensors or alternative methods to determine visibility conditions.
- **Software Updates:** The software controlling the wiper system, and the data it provides, can be updated over the air (OTA) to improve performance or add new features.

This table provides a general overview. For a specific vehicle, you'd need to consult the vehicle's technical documentation (e.g., wiring diagrams, CAN bus specifications) to get the precise details.

3.4 Functionality of Other Items Required by This Item

Clause: 5.4.2 d)

Here's a breakdown of the Wiper System's dependencies on other vehicle systems, including the requested details:

(1) Source System	(2) Data/Function Required	(3) Timing Dependency	(4) Failure Consequence
Body Control Module (BCM)	**Wiper Switch Position (e.g., Off, Low, High, Intermittent, Wash)**	Continuous (as switch position changes)	Loss of wiper control. Wipers may remain in a default state (e.g., off) or operate at a default speed.
Body Control Module (BCM)	**Rain Sensor Data (if equipped) - Rain Intensity Level**	Continuous (as rain intensity changes)	Wiper system may not automatically adjust speed based on rain. Wipers may operate at a default intermittent speed or not at all.
Vehicle Speed Sensor (VSS) / Powertrain Control Module (PCM) / ABS Module	**Vehicle Speed**	Continuous (as vehicle speed changes)	Wiper speed may not adjust based on vehicle speed (e.g., slower at low speeds, faster at high speeds). Wipers may operate at a default speed.
Powertrain Control Module (PCM) / Engine Control Unit (ECU)	**Engine Running Status (e.g., Engine On/Off)**	Startup and Continuous (while engine is running)	Wipers may not operate if the engine is off (depending on system design). Wipers may be disabled to conserve battery power.
Power Distribution Module (PDM) / Fuse Box	**Power Supply (e.g., 12V)**	Continuous (when wipers are intended to operate)	Wipers will not function.
Body Control Module (BCM)	**Washer Fluid Level (if equipped)**	Continuous (when washer fluid is used)	Washer fluid spray may not be activated.
Body Control Module (BCM)	**Headlight Status (e.g., Headlights On/Off)**	Continuous (when headlights are on)	Headlight-linked wiper functions (e.g., automatic wipe when headlights are turned on) may not work.
HVAC System (if equipped with auto-defog)	**Defogging Request (e.g., Defog On/Off)**	Continuous (when defogging is requested)	Wipers may not activate automatically to clear condensation from the windshield.

| **CAN Bus (if applicable)** | **Communication with other modules (e.g., BCM, PCM, ABS)** | Continuous (for data exchange) | Loss of communication can lead to incorrect wiper operation, failure to respond to commands, or complete system failure. |

| **Steering Angle Sensor (if equipped with speed-sensitive wipers)** | **Steering Angle** | Continuous (when speed-sensitive wipers are active) | Wiper speed may not adjust based on steering angle. |

| **Parking Brake Status (if equipped with wiper disable when parking brake is engaged)** | **Parking Brake Status (e.g., Engaged/Disengaged)** | Continuous (when parking brake is engaged) | Wipers may not operate when the parking brake is engaged. |

| **Door Lock Status (if equipped with wiper activation when doors are unlocked)** | **Door Lock Status (e.g., Locked/Unlocked)** | Continuous (when doors are unlocked) | Wipers may not activate when the doors are unlocked. |

3.5 Allocation and Distribution of Functions

Clause: 5.4.2 e)

Safety-Critical Function Allocation in a Wiper System

The allocation of safety-critical functions between hardware and software in a wiper system is crucial for ensuring reliable and safe operation. This allocation aims to distribute responsibilities to minimize single points of failure and leverage the strengths of each domain. Here's a breakdown:

System Overview:

The wiper system typically consists of:

- **Wiper Motor:** The actuator responsible for moving the wiper blades.
- **Wiper Blades:** The physical components that clear the windshield.
- **Wiper Linkage:** The mechanical system connecting the motor to the blades.
- **Wiper Control Module (WCM):** The electronic control unit (ECU) that manages the wiper system.
- **Sensors:** Rain sensor, vehicle speed sensor, and potentially others.
- **Power Supply:** Provides electrical power to the system.
- **User Interface:** Wiper stalk or buttons for user input.

Function Allocation Table:

| Safety-Critical Function | Hardware Component(s) | Software Component(s) | Responsible Entity (e.g., "WCM Software") | Notes

3.6 Operational Scenarios Impacting Functionality

Clause: 5.4.2 f)

Here are some realistic operational scenarios that challenge the "Wiper System" safety, along with their descriptions:

1. Scenario: Heavy Snowfall During Night Driving

***(1) Scenario Conditions:*

- Heavy snowfall accumulating rapidly on the windshield.
- Nighttime conditions, reducing visibility further.
- Driving at highway speeds (e.g., 60 mph / 96 km/h).
- Ambient temperature near or below freezing (e.g., 30°F / -1°C).
- Headlights and other vehicle lights are on.

***(2) Expected Item Behaviour:*

- Wiper blades struggle to clear the windshield effectively due to the volume and density of the snow.
- Snow accumulates on the windshield, reducing visibility to near zero.
- Wiper motor may experience increased load and potential overheating if used continuously at high speed.
- Ice may form on the wiper blades and windshield, reducing their effectiveness and potentially damaging the blades.
- Headlights may be partially obscured by snow, reducing their effectiveness.

***(3) Safety Mechanisms Activated:*

- **Automatic Wiper Speed Adjustment:** The wiper system automatically increases the wiper speed to the highest setting (or a pre-defined maximum) to attempt to clear the snow.
- **Heated Windshield/Wiper Blades (if equipped):** The heated elements activate to melt ice and snow, improving wiper performance.

- **Windshield Washer Fluid Activation:** The driver may repeatedly activate the washer fluid to help melt snow and improve visibility. The fluid may be a winter-blend with antifreeze.
- **Driver Alert/Warning:** The vehicle may provide a visual or audible warning to the driver to reduce speed and/or pull over if visibility is severely impaired.
- **Reduced Speed/Adaptive Cruise Control Adjustment:** The driver may manually reduce speed or the adaptive cruise control system may automatically reduce speed to maintain a safe following distance.
- **Emergency Braking System (if equipped):** In extreme cases of reduced visibility and a potential collision, the emergency braking system may activate to mitigate the impact.

2. Scenario: Sudden Rainstorm During Overtaking Maneuver

***(1) Scenario Conditions:*

- Clear, dry road conditions.
- Vehicle is in the process of overtaking another vehicle on a two-lane highway.
- Sudden onset of heavy rainfall, drastically reducing visibility.
- Vehicle speed is relatively high (e.g., 65 mph / 105 km/h).

***(2) Expected Item Behaviour:*

- Visibility is immediately and severely reduced due to the sudden downpour.
- The driver's reaction time is critical to maintain control and avoid a collision.
- The wiper system is immediately engaged, but may not be able to clear the windshield fast enough initially.
- The road surface becomes slick, increasing the risk of hydroplaning.

***(3) Safety Mechanisms Activated:*

- **Automatic Wiper Activation:** Rain sensors automatically detect the rain and activate the wipers, starting at a low speed and increasing as the rain intensifies.
- **High Wiper Speed:** The driver may manually or the system automatically select the highest wiper speed to clear the windshield as quickly as possible.
- **Windshield Washer Fluid Activation:** The driver may use the washer fluid to clear initial droplets and improve visibility.
- **Reduced Speed/Adaptive Cruise Control Adjustment:** The driver should immediately reduce speed to maintain control and increase following distance. Adaptive cruise control may automatically reduce speed.

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- **Stability Control System (ESC/ESP):** The ESC system may intervene to help maintain vehicle stability if the vehicle begins to skid or hydroplane.
- **Emergency Braking System (if equipped):** In extreme cases of reduced visibility and a potential collision, the emergency braking system may activate to mitigate the impact.
- **Hazard Lights Activation:** The driver may activate hazard lights to alert other drivers of the hazardous conditions.

3. Scenario: Wiper Blade Failure During Highway Driving

***(1) Scenario Conditions:*

- Driving on a highway at high speed (e.g., 70 mph / 113 km/h).
- Light rain or drizzle, requiring wiper operation.
- One or both wiper blades fail (e.g., blade detaches, rubber tears, or motor failure).

***(2) Expected Item Behaviour:*

- Visibility is immediately and significantly impaired, especially if the failure occurs on the driver's side.
- The driver's ability to see the road and react to hazards is severely compromised.
- The remaining wiper blade(s) may be overloaded or ineffective.
- The driver may experience panic or difficulty maintaining control.

***(3) Safety Mechanisms Activated:*

- **Driver Reaction:** The driver must immediately react to the loss of visibility.
- **Reduced Speed/Emergency Stop:** The driver should immediately reduce speed and attempt to pull over to a safe location.
- **Hazard Lights Activation:** The driver should activate hazard lights to alert other drivers.
- **Lane Keeping Assist (if equipped):** The lane keeping assist system may provide some assistance in maintaining lane position, but its effectiveness is limited by the reduced visibility.
- **Emergency Braking System (if equipped):** In extreme cases of reduced visibility and a potential collision, the emergency braking system may activate to mitigate the impact.
- **Driver Assistance Systems Deactivation:** The driver may need to manually deactivate driver assistance systems (e.g., lane keeping assist, adaptive cruise control) if they are not functioning correctly due to the wiper failure.

4. Scenario: Sun Glare and Dirty Windshield

***(1) Scenario Conditions:*

- Driving into the setting sun, creating significant glare.
- Windshield is dirty with road grime, dust, or insect residue.
- Driving at highway speeds (e.g., 65 mph / 105 km/h).

***(2) Expected Item Behaviour:*

- Sun glare significantly reduces visibility, making it difficult to see the road and other vehicles.
- The dirty windshield exacerbates the glare, scattering light and further reducing visibility.
- The driver may struggle to see lane markings, traffic signals, and other hazards.
- The wipers may smear the dirt and grime, further reducing visibility if used without washer fluid.

***(3) Safety Mechanisms Activated:*

- **Driver Reaction:** The driver must immediately react to the reduced visibility.
- **Sun Visor Deployment:** The driver should deploy the sun visor to block the sun.
- **Windshield Washer Fluid Activation:** The driver should use the washer fluid to clean the windshield.
- **Wiper Operation:** The driver should use the wipers to clear the windshield after applying washer fluid.
- **Reduced Speed:** The driver should reduce speed to compensate for the reduced visibility.
- **Increased Following Distance:** The driver should increase following distance to allow for more reaction time.
- **Headlights Activation:** The driver may activate headlights to improve visibility for themselves and other drivers.

These scenarios highlight the critical role of the wiper system in vehicle safety and the various mechanisms that are activated to mitigate risks in challenging driving conditions. The effectiveness of these mechanisms depends on the design and functionality of the wiper system, as well as the driver's awareness and response.

4. Conclusion

I confirm that this document satisfies ISO 26262-3:2018 Clause 5.4. It provides sufficient detail to proceed to HARA (Clause 6) and Functional Safety Concept (Clause 7).

5. Approvals

Role	Name	Signature	Date
Safety Engineer	*[Placeholder]*	*[Placeholder]*	*[Placeholder]*
System Architect	*[Placeholder]*	*[Placeholder]*	*[Placeholder]*
Project Manager	*[Placeholder]*	*[Placeholder]*	*[Placeholder]*